



Cite this: DOI: 10.1039/c8tb02331j

Received 4th September 2018,
Accepted 24th October 2018

DOI: 10.1039/c8tb02331j

rsc.li/materials-b

Multifunctional cellulose-based hydrogels for biomedical applications

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In recent decades, cellulose has been extensively investigated due to its favourable properties, such as hydrophilicity, low-cost, biodegradability, biocompatibility, and non-toxicity, which makes it a good feedstock for the synthesis of biocompatible hydrogels. The plentiful hydrophilic functional groups (such as hydroxyl, carboxyl, and aldehyde groups) in the backbone of cellulose and its derivatives can be used to prepare hydrogels easily with fascinating structures and properties, leading to burgeoning research interest in biomedical applications. This review focuses on state-of-the-art progress in cellulose-based hydrogels, which covers from their preparation methods (including chemical methods and physical methods) and physicochemical properties (such as stimuli-responsive properties, mechanical properties, and self-healing properties) to their biomedical applications, including drug delivery, tissue engineering, wound dressing, bioimaging, wearable sensors and so on. Moreover, the current challenges and future prospects for cellulose-based hydrogels in regard to their biomedical applications are also discussed at the end.

1. Introduction

Hydrogels are three-dimensional (3D) polymeric materials, which can absorb and release a large amount of water or biological fluids in a reversible manner.¹ Some of them act in response to specific environmental stimuli including pH, temperature, light, ions, and magnetic and electric fields, *etc.*^{1,2} Such stimuli-sensitive behaviour makes hydrogels appealing for the design of “smart” devices, and applicable in biomedical materials, hygiene, water purification, agriculture, and horticulture, *etc.*^{3–5} According to the polymer source, hydrogels can be divided into natural and synthetic polymer-based hydrogels. Natural polymer-based hydrogels are derived from various

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natural hydrophilic polymers, such as cellulose, gelatin, peptides, chitosan, agar-agar, alginate, hyaluronic acid, and some of their derivatives,^{6–8} while synthetic polymer-based hydrogels often relate to poly(ethylene glycol) (PEG), poly(acrylic acid) (PAA), polyacrylamide (PAAm), poly(vinyl alcohol) (PVA), and poly(*N*-isopropylacrylamide) (PNIPAAm), and their copolymers.^{9–12} Thanks to their essential biocompatibility, biodegradability and tissue-mimicking consistency characteristics, natural polymer-based hydrogels are drawing tremendous popularity in biomedical daily-care, pharmaceutical smart wound dressing, ionic skins, healthcare monitoring, and human-machine interfaces.^{13–17}

Cellulose is a fascinating natural polymer composed of β -(1 $\rightarrow$ 4) linked D-glucose units, which has been considered as an inexhaustible feedstock for the synthesis of multifunctional hydrogels.^{4,7,18} The plentiful hydrophilic functional groups, including hydroxyl, carboxyl, and aldehyde groups, in the backbone of cellulose and its derivatives can be used to prepare hydrogels easily with fascinating structures and properties, making them promising biomaterials for various biomedical applications including drug delivery,^{19,20} tissue engineering,^{21,22} wound dressing,^{23–25} bioimaging,^{26–28} and wearable epidermal sensors,^{16,29,30} etc.

Cellulose-based hydrogels possess some attractive properties that make them suitable for biomedical applications: (i) as a naturally-occurring polysaccharide produced by plants or bacteria, cellulose has intrinsic biocompatibility, biodegradability, non-toxicity, and low cost, endowing cellulose-based hydrogels with sustainable and biodegradable properties; (ii) it is well known that there are numerous intermolecular and intramolecular hydrogen bonds in the structure of cellulose, thus giving excellent mechanical performance to the cellulose network, which can be used as reinforcement materials of hydrogels.³¹ Cellulose-based hydrogels with high strength and

high toughness are promising candidates for defective bone repair, especially load-bearing soft tissues such as cartilage;^{22,32} (iii) by crosslinking cellulose with stimuli-sensitive polymers, stimuli-responsive cellulose hydrogels can be obtained.^{33,34} These “smart” hydrogels could not only protect loaded drugs, proteins and peptides from the environment, the release of them can be accelerated or slowed down according to the shrinking or swelling of polymeric chains leading to “on-demand” remote control and “on-off” switchable control drug delivery, which are important in various biomedical fields;^{19,35} (iv) cellulose nanoparticles (NPs), such as cellulose nanocrystals (CNCs) that are extracted by acid hydrolysis from native cellulose, can be used as fillers to reinforce polymer hydrogels due to their excellent mechanical properties, resulting in hydrogels with mechanical robust and rapid self-healing properties, which are promising candidates for wearable strain sensors.^{29,30}

The fascinating properties of cellulose-based hydrogels together with their inherent biocompatibility have aroused extensive research interest for various biomedical applications. Since the classification and dissolution of native cellulose and cellulose derivatives and the design and fabrication approaches of cellulose-based hydrogels have been described in detail in previous reviews,^{4,18,36} in this review, we will focus on the fascinating properties and biomedical applications of cellulose-based hydrogels. We first discuss their synthesis methods including chemical methods and physical methods. Then, we highlight their unique properties, such as stimuli-responsive properties, mechanical properties, and self-healing properties. Afterwards, we emphasize their various biomedical applications, such as drug delivery, tissue engineering, wound dressing, bioimaging, and wearable sensors (Fig. 1). Finally, we elaborate on the current challenges and future prospects of cellulose-based hydrogels for new biomedical applications.



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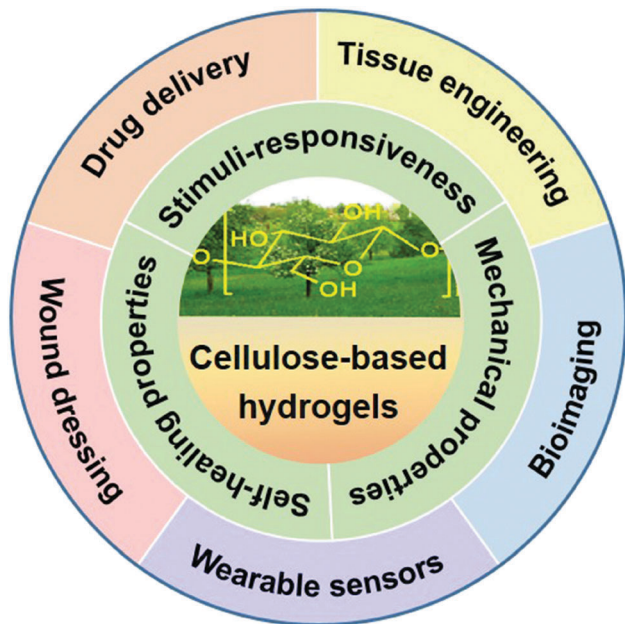


Fig. 1 An overview of the properties and biomedical applications of cellulose-based hydrogels in the biomedical field.

2. Synthesis of cellulose-based hydrogels

Generally, the fabrication methods of cellulose-based hydrogels can be divided into two categories, chemical methods and physical methods. The corresponding hydrogels are called chemical gels and physical gels, respectively. Chemical methods involve the formation of covalent bonds between polymers *via* free radical polymerization,^{37,38} radical copolymerization,³⁹ or graft copolymerization,⁴⁰ *etc.*, which leads to permanent junctions in the polymer network. On the contrary, physical gels are held together by reversible physical interactions (noncovalent bonds) such as polymer chain entanglements, van der Waals forces, ionic interactions, hydrogen bonding, and crystallite associations.^{4,18} In hydrogel networks, the chemical cross-linking points achieved by covalent bonds can maintain the elasticity of the hydrogels, while the reversible physical cross-linking points stemming from noncovalent bonds allow hydrogels to recover or self-heal after large deformation or disruption.⁴¹ For chemical methods, a functionalized chemical cross-linker is usually required during the gelation, since the cross-linkers can form covalent bonds that link cellulose or its derivatives to different polymer molecules in a 3D hydrophilic network. Widely used cross-linkers are *N,N'*-methylenebisacrylamide (MBA),^{38,40} epichlorohydrin (ECH),^{27,42} glutaraldehyde,⁴² polyethylene glycol diacrylate (PEGDA),⁴³ citric acid,⁴⁴ and 1,3-diaminopropane,⁴⁵ *etc.* For example, CNC-PAAm composite hydrogels with excellent mechanical viscoelastic properties were obtained by immersing a CNC skeleton into a AAm monomer solution, and *in situ* polymerization occurred in the presence of cross-linker PEGDA and initiator potassium persulfate.⁴³ Under stress, hydrogen bonds between CNC and PAAm as well as inside

the CNC skeleton acted as sacrificial bonds to dissipate energy, while the covalently cross-linked PAAm chains bind the network together by providing adhesion to CNC and thereby suppress catastrophic crackle propagation. The synergistic effects of the CNC and the elastic PAAm network endowed the hydrogels with excellent mechanical properties, which could withstand up to 181 kPa of tensile stress, 1.01 MPa of compressive strength, and 1392% elongation at break with a fracture energy of 2.82 kJ m⁻². Moreover, the hydrogels recovered more than 70% elasticity after eight loading-unloading cycles, revealing excellent fatigue resistance. Depth-sensing instrumentation for indentation testing corroborated that the introduction of CNC could simultaneously improve the hardness and elasticity of the composite hydrogels by 500% in comparison with PAAm hydrogels.

Chemical synthesis methods can also be realized under appropriate doses of irradiation, such as ultraviolet (UV),⁴⁶ gamma ray,⁴⁷ and electron beam (EB) radiation.⁴⁸ By avoiding the use of toxic chemical cross-linkers, the radical cross-linking method can achieve cellulose-based hydrogels with higher purity, and can find more applications in the food, drug, and pharmaceutical industries.¹⁸ Yang *et al.* prepared a mechanically tough and elastic CNC-PEG nanocomposite hydrogel by directly mixing CNC, PEG, and 2-hydroxy-1-[4-(2-hydroxyethoxy)-phenyl]-2-methyl-1-propanone aqueous suspension; after degassing, free radical polymerization was initiated by UV irradiation (20 W, 365 nm, 20 min) at room temperature without any crosslinking agents.⁴⁶ Compared with neat PEG hydrogels, the composite hydrogels exhibited higher strengths and flexibilities at a CNC loading of 1.2% (v/v), where the modulus, fracture stress, and fracture strain were enhanced 3.48, 5, and 3.28 times over the PEG hydrogels, respectively. Oscillatory shear tests indicated the composite hydrogels were more viscous than PEG hydrogels and were efficient at energy dissipation due to the reversible interactions between CNC and PEG chains. It was proposed that the strong gel viscoelastic behaviour and the mechanical reinforcement were related to the “filler network”, where the temporary interactions between CNC and PEG interfered with the covalent cross-links of PEG.

Unlike chemical methods, there is no covalent bond formation in the process of physical methods, in which cross-linked networks are formed by physical interactions between the polymers. The freeze-thaw technique is the commonly used method to prepare physical hydrogels.^{49–51} Besides, physical gels can also be obtained by ultrasonic treatment,⁵² or by directly stirring the mixture of precursors.⁵³ Wang *et al.* prepared a composite hydrogel from PVA and carboxymethyl cellulose (CMC) *via* the freeze-thaw technique.⁴⁹ PVA and CMC aqueous solution were mixed with different ratios, and frozen at –20 °C for 6 h, followed by thawing at room temperature for 1 h. After repeating the freeze-thaw cycle five times, PVA/CMC hydrogels were obtained. The obtained PVA/CMC composite hydrogels possessed a sponge-like and porous structure, and the maximum swelling ratio was improved to 1437%, whereas the swelling ratio of the neat PVA hydrogels was 416%. The higher water uptake ability of composite hydrogels may be due to the increasing hydrophilicity from CMC. In addition, the cross-linking density

in the hydrogel decreased with the decreasing of PVA, which benefited the permeation of water molecules, leading to the improvement of swelling ability. Similarly, PVA/CNC hydrogels were also obtained by using the freeze-thaw technique, and the compressive properties of the hydrogels were found to be improved by the addition of CNC.⁵⁰

Importantly, the combination of chemical and physical cross-linking in a single cellulose-based hydrogel system has emerged as a promising strategy to prepare hydrogels with enhanced mechanical performance, good conductivity, reliable self-healing capability, and high strain sensitive properties, which show great potential in the application of wearable strain sensors.^{29,30} The required mechanical toughness and excellent mobility properties for wearable strain sensors are two contradictory characteristics, which are derived from their inherent compromise between dynamic cross-links for healing and steady cross-links for mechanical strength.⁵⁴ Inspired by biological soft tissues with hierarchical network structures, CNC-based hydrogels with “soft and hard” hierarchical network structures were constructed by combining smooth stress-transfer and reversible coordination bonds (Fig. 2).²⁹ In this system, covalent cross-linked PVA and poly(vinylpyrrolidone) (PVP) served as a “soft” homogeneous polymer network (Fig. 2A and D), and a “hard” Fe^{3+} cross-linked CNC network (Fig. 2B and E) with dynamic CNC- Fe^{3+} ionic coordination bonds was interconnected as nanoreinforcing domains. These synergistic “soft and hard” hierarchical network structures (Fig. 2C) were demonstrated to enable functional network hydrogels to achieve excellent mechanical performance, reliable self-healing capability, and good self-recovery properties. Similarly, a tough and self-healing cellulose-based hydrogel was constructed by combining multiple coordination bonds among tannic acid (TA)-coated CNCs (TA@CNC), PAA chains, and Al^{3+} ions in a covalent PAA network.³⁰ The incorporated TA@CNC acted as a

dynamic connected bridge in the hierarchically porous network that was mediated by multiple reversible coordination bonds among TA, CNC, and Al^{3+} ions, endowing this gel with superior mechanical properties, excellent recoverability and electrical self-healing properties. Due to the presence of catechol groups from incorporated TA, these gels display durable and repeatable adhesiveness to human skin tissue without an inflammatory response, thus they could be employed as flexible strain sensors adhered directly on human skin.

Besides the abovementioned methods, many new techniques such as hydrogel foaming,⁵⁵ the enzymatic synthesis method,^{56,57} solid-liquid interface contact technology and the layer by layer method⁵⁸ have been developed to prepare unique hydrogels. For example, multi-layered tubular cellulose (MLTC) hydrogels were constructed *via* the fast contact of the solid-liquid interface in NaOH/urea aqueous solution by using agarose as a template to control the structure of the hydrogels.⁵⁸ For the preparation of MLTC hydrogels, an agarose gel rod (AGR) was immersed into acetic acid (AA) solution for 1–10 min, and then the AA-loaded AGR was contacted with cellulose solution on the solution-solid interface for the same time to prepare the first cellulose layer. Subsequently, the gel rod with the first layer was immersed again in AA, thus the new AA-loaded gel rod could be used to form the second layer. By repeating this process, onion-like MLTC hydrogels with smooth cellulose layers were obtained, and the cellulose layer thickness could be controlled in a wide range of 36–305 μm simply by adjusting the experimental parameters (including the cellulose concentration, the diameter of the AGR core, the amount of loaded AA and the soaking time). The stiff and hydrophilic cellulose chains contributed to support the thin wall of pores, leading to the regular macroporous structure. The prepared hydrogels exhibited a compressive strength higher than 0.3 MPa, whereas the strain was more than 55%. Moreover, the multi-layered architecture

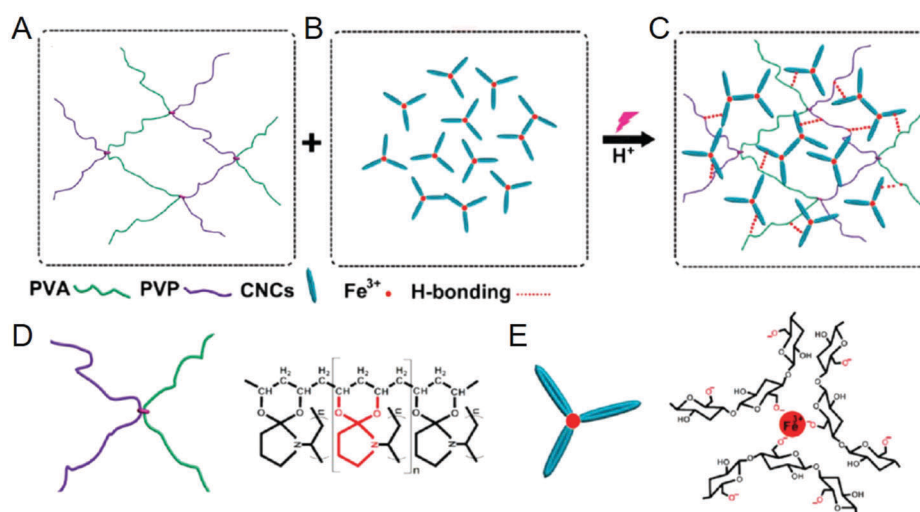


Fig. 2 Schematic illustration of synergistic “soft” and “hard” hierarchical network structures: (A) “soft” homogeneous PVA–PVP networks; (B) “hard” CNCs as reinforcing domains bound together by dynamic CNC– Fe^{3+} binding units; and (C) synergistic “soft and hard” hierarchical network structures containing extensive interchain hydrogen bonding. (D and E) The chemical formula of homogeneous PVA–PVP networks (D) and CNC– Fe^{3+} binding units (E). Reproduced from ref. 29 with permission from the American Chemical Society.

could be well-maintained in different solvents (including ethanol, acetone, dimethylacetamide and 10 wt% NaOH aqueous solution), and exhibited good capacity for cell proliferation using L929 cells as seed cells. These MLTC hydrogels with high compressive strength, good solvent resistance, and good capacity for cell proliferation have potential for various applications such as tissue engineering scaffolds.

Moreover, 3D printing technology has been introduced to prepare cellulose-based hydrogel scaffolds from an ink of sodium alginate (SA) and gelatin reinforced with CNCs.⁵⁹ The CNCs provided favourable rheological properties required for 3D printing. The printed scaffolds were crosslinked sequentially *via* covalent and ionic reactions resulting in dimensionally stable hydrogel scaffolds with pore sizes of 80–2125 μm and nanoscale pore wall roughness favourable for cell interaction. By optimising the printing pathways, scaffolds (Z axis up to 20 mm) with uniform and gradient pore structures were achieved, which was promising for the customization of biomedical scaffolds with respect to size, shape, pore structure and orientation, useful for tissue repair and regeneration.

3. Properties of cellulose-based hydrogels

The abundant hydrophilic functional groups (such as hydroxyl, carboxyl, and aldehyde groups) in the backbone of cellulose and its derivatives can be used to prepare hydrogels easily with fascinating structures and properties. In this section, we will highlight the favourable properties of cellulose-based hydrogels in detail, such as stimuli-responsive properties, mechanical properties and self-healing properties.

3.1 Stimuli-responsive cellulose-based hydrogels

Stimuli-responsive cellulose-based hydrogels have the unique property of undergoing abrupt volume changes from their collapsed or swollen states in response to environmental stimuli, making them possess great potential in biomedical applications, especially for drug delivery.^{4,6,19} The environmental stimuli include pH, temperature, redox, light, solvent composition, electric and magnetic fields, and even biological/biochemical ones such as glucose (Fig. 3). Generally, these stimuli-responsive cellulose-based hydrogels can be divided into two categories, single and multi-stimuli-responsive hydrogels, which are summarized in Table 1 and discussed in detail below.

3.1.1 Single stimuli-responsiveness

pH-Responsive cellulose-based hydrogels. pH-sensitive hydrogels can change their volume by alteration of the pH of the external environment, which can be used to control drug release. Different from normal tissue and blood that have a pH value of 7.4, most pathological tissues exhibit lower pH values. For example, the extracellular pH in tumour micro-environments is 6.5–6.8, and the pH of intracellular endosome and lysosome is decreased to 5.0–5.5.⁸⁸ The pH of the stomach is even harsher, which can be decreased to below 3.0 due to the secretion of stomach acid. Therefore, ideal hydrogel drug

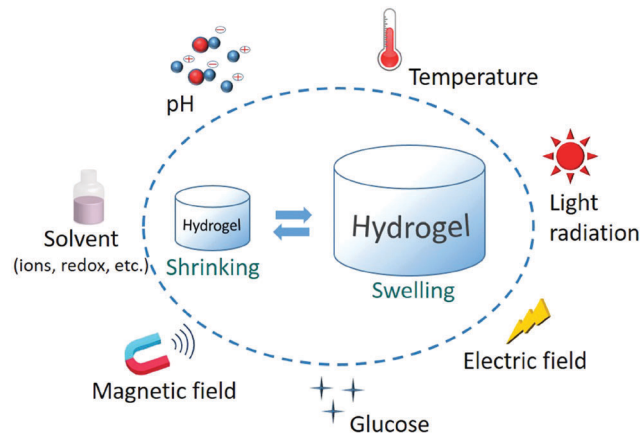


Fig. 3 Stimuli-responsive property of cellulose-based hydrogels.

delivery systems should not only be biocompatible and biodegradable, but also pH-responsive to protect the drugs and release them at lesion locations. For example, the pH-sensitive polymer CMC is an anionic and water-soluble cellulose derivative obtained by introducing $-\text{CH}_2\text{COOH}$ groups into cellulose molecular chains. In acidic media, the formed hydrogen bonds in CMC networks render the networks more hydrophobic, resulting in a collapsed state. On the contrary, in alkaline media, the acidic carboxyl groups on the CMC chains are ionized into carboxylate groups ($-\text{COO}^-$), leading to the swelling of CMC-based hydrogels due to the increase of electrostatic repulsion between these charged groups and increase of osmotic pressure. Based on this, pH-sensitive CMC-based hydrogels were developed.^{33,60–62}

Zare-Akbari *et al.* synthesized ionic cross-linked hydrogel beads based on CMC and ZnO NPs by using Fe^{3+} ions as a physical crosslinking agent.⁶¹ The prepared hydrogel beads were shown to possess pH-sensitive swelling behaviour by immersing in phosphate buffer solution (PBS) with pH 1.2, 6.8 and 7.4 at room temperature to mimic different physiological environments. It was found that the swelling degree at pH 7.4 and 6.8 was about three times than that at pH 1.2. At pH 6.8 and 7.4, carboxyl groups in CMC were dissociated in aqueous media and converted to negatively charged carboxylate ions, leading to higher electrostatic repulsion and thus, water penetrating into the hydrogel matrix. The pH-sensitive drug release behaviour was demonstrated by using propranolol hydrochloride as a model drug, suggesting that the hydrogel beads might be a candidate for controlled delivery of drugs.

Besides, many polymers containing weakly acidic or basic groups that easily undergo hydrolysis or protonation, have been developed to prepare pH-sensitive hydrogels with cellulose, such as gelatin,⁴² PAA,⁴⁸ hyaluronic acid (HA),⁶⁵ and poly(*N,N*-dimethylaminoethyl methacrylate) (PDMAEMA).⁷⁵ Generally, the cellulose-based hydrogels containing anionic polymers (such as CMC, PAA, and HA) swell more at high pH due to the ionization of carboxyl groups,^{48,61,65} whereas the hydrogels containing cationic polymers such as PDMAEMA swell more at low pH.⁷⁵ The pH sensitivity of hydrogels containing PDMAEMA is

Table 1 Stimuli-responsive cellulose-based hydrogels and their biomedical applications

Category	Hydrogels	Stimuli	Applications	Ref.
Single-responsiveness	CMC/PAA	pH	Oral insulin delivery	60
	CMC/ZnO hydrogel beads	pH	Propranolol hydrochloride delivery	61
	CMC/graphene oxide	pH	Doxorubicin delivery	33 and 62
	CNC–gelatin hydrogels	pH	Theophylline delivery	42
	CNF–gelatin hydrogels	pH	NR	63
	CNFs–based polyion complex hydrogels	pH	Doxorubicin delivery	19
	Microfibrillated cellulose–PVA–borax hydrogels	PH	NR	64
	BC-g-PAA	pH	Bovine serum albumin delivery	48
	HEC/hyaluronic acid	pH	Isoliquiritigenin delivery	65
	Graphene oxide–cellulose/PVA hydrogels	pH	NR	66
	PNIPAAm–CNC hybrid hydrogels	Temperature	Metronidazole delivery	67
	CNCs–PNIPAAm	Temperature	Cell encapsulation	68
	TO–CNFs/PNIPAAm	Temperature	NR	38
	CNC–poly(NIPAAm-co-DMAEMA)	Temperature	Cell encapsulation	69
	CNC/PVCL	Temperature	NR	70
	CNF/tetra-arm-PEG	Temperature	NR	53
	MC/CNC	Temperature	NR	71
	PNIPAAm/CNWs	Temperature	NR	34
	CNC–imidazole hydrogels	CO ₂	NR	72
	CNC/1-(3-aminopropyl)imidazole	CO ₂	NR	73
Multi-responsiveness	Quaternized cellulose/CMC	pH, salt	NR	74
	PDMAEMA/CNW/carboxymethyl chitosan	pH, temperature	Flutamide delivery	75
	CNW–PAAm–PNIPAAm	pH, temperature	NR	76
	TO–CNF/PNIPAAm	pH, temperature	Methylene blue delivery	77
	BC/acrylic acid	pH, temperature	Bovine serum albumin delivery	78
	BC nanofibers/sodium alginate	pH, electron	Ibuprofen delivery	79
	Cellulose/4-vinyl-phenylboronic acid	pH, glucose	Insulin delivery	80
	Cellulose acetoacetate–cystamine dihydrochloride	pH, redox	Rhodamine B delivery	81
	CEC–TPH/PEG–DA hydrogels	pH, redox	Doxorubicin delivery and 3D cell culture	82
	CMC/MC/Fe ₃ O ₄	Magnetic, electric	Curcumin delivery	83
	Cotton fibers–spiropyran–NIPAAm hydrogels	Temperature, light (sunlight or white light)	NR	84
	CNF–PNIPAAm–clay bilayer nanocomposite hydrogels	Temperature, near infrared light	Soft actuators	37
	Methacrylated CMC/PNIPAAm	pH, temperature, redox	Lysozyme delivery	85
	Dialdehyde cellulose-based hydrogels	pH, redox, UV illumination	NR	86
	CNC self-assembled with various monomers	pH, temperature, solvent	NR	87

Notes: NR: not reported; CMC: carboxymethyl cellulose; CNCs: cellulose nanocrystals; CNFs: cellulose nanofibrils; HEC: hydroxyethyl cellulose; CNWs: cellulose nanowhiskers; MC: methyl cellulose; BC: bacterial cellulose; (2,2,6,6-tetramethylpiperidin-1-yl)oxyl-oxidized cellulose nanofibers (TO–CNFs); PAA: poly(acrylic acid); PVA: poly(vinyl alcohol); PNIPAAm: poly(*N*-isopropylacrylamide); PEG: poly(ethylene glycol); PAAm: polyacrylamide; poly(NIPAAm-co-DMAEMA): poly(*N*-isopropylacrylamide-co-*N,N'*-dimethylaminoethylmethacrylate); PVCL: poly(*N*-vinylcaprolactam); PDMAEMA: poly(*N,N*-dimethylaminoethyl methacrylate); CEC–TPH: carboxyethyl cellulose-*g*-dithiodipropionate dihydrazide; PEG–DA: dibenzaldehyde-terminated PEG.

attributed to the protonation of the tertiary amine groups, which become protonated in a low pH medium and are rendered hydrophilic, whereas they become hydrophobic on deprotonation at high pH. For amphoteric polymers, such as gelatin, they have both carboxyl and amino groups on the side chains with an isoelectric point of pH 4.9; the swelling behaviours at different pH values depend on whether gelatin is acting as an acid or as a base.⁴² When the pH of the medium is lower than 4.9, the gelatin network will form a cationic gel, while an anionic gel can be obtained when the pH is higher than 4.9. In a word, the increase of ionization could cause the hydrogel to swell significantly and release the drug by changing the pH of the external environment. Therefore, we can choose the appropriate polymers to construct pH-sensitive cellulose-based hydrogel drug delivery systems depending on the pH of lesion locations.

Thermo-responsive cellulose-based hydrogels. For biomedical applications, a thermo-responsive (or temperature-responsive) system is regarded as the most capable trigger system because human body temperatures are around 36.5–37.5 °C regardless of the functions of the organs, tissues and cells.⁸⁹ Many thermo-sensitive polymers have been utilized to construct temperature-responsive cellulose-based hydrogels. These thermosensitive polymers can be categorized into negatively and positively thermosensitive polymers:^{2,88} (1) negatively thermosensitive polymers, such as PNIPAAm, possess a lower critical solution temperature (LCST), which can form a gel when $T > \text{LCST}$ and return into solution when $T < \text{LCST}$; (2) positively thermo-sensitive polymers (such as polybetaines) have an upper critical solution temperature (UCST), which exhibit the opposite behaviour to negatively thermosensitive polymers, and they are a gel when $T < \text{UCST}$ and are soluble when $T > \text{UCST}$.

Negatively thermosensitive hydrogels are frequently used in biomedical fields. By changing the temperature, the release of drugs from thermosensitive hydrogels could be accelerated or slowed down according to the shrinking or swelling of polymeric chains, which is called “on-demand” remote control as well as “on-off” switchable control drug delivery.¹

PNIPAAm is one of the most representative thermosensitive amphiphilic polymer materials for the construction of temperature-responsive cellulose-based hydrogels.^{34,38,67–69} Thérien-Aubin *et al.* prepared a nanofibrillar hydrogel from rod-shaped CNCs functionalized with NIPAAm and *N,N'*-dimethylaminoethylmethacrylate.⁶⁹ The prepared polymer-modified CNCs (mCNCs) display inverse thermo-reversible properties, *i.e.* the mCNC suspension exhibited low extinction and was clear at room temperature, whereas the suspension displayed a sharp increase in extinction and turned turbid when heated to 37 °C (above the lower critical aggregation temperature of mCNCs). Moreover, upon heating, the viscous behaviour was found to have a 10-fold increase in the storage modulus (G'), which was higher than the loss modulus (G''), indicating the formation of a hydrogel. The prepared mCNCs were used for cell encapsulation and culture of fibroblasts and T-cells. As shown in Fig. 4, the mCNCs were mixed with cells in a cell culture medium at room temperature, then a cell-laden hydrogel was obtained by heating the mixture to 37 °C. Upon cooling the hydrogel to room temperature, the hydrogel was dissociated, leading to cell release. It was found that the temperature-responsive mCNCs could significantly promote the proliferation of T cells (suspension cells) with a proliferation index of 2–4.5 after one day of cell culture. Due to the low cytotoxicity, biocompatibility and tunable properties, the temperature-responsive mCNCs might be used as injectable gels and as an artificial extracellular matrix for 3D cell culture.

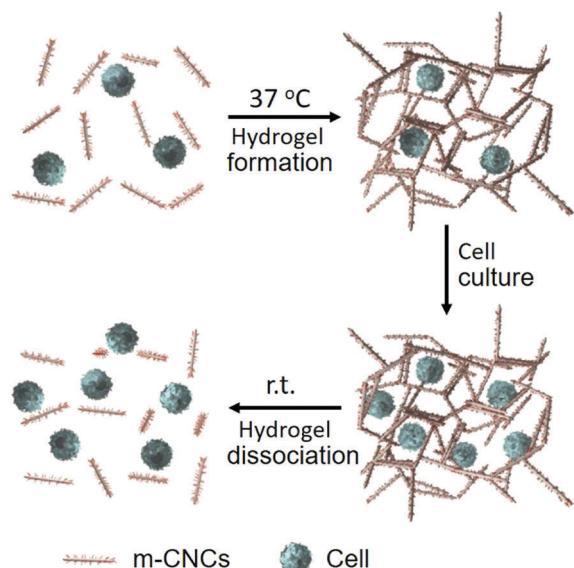


Fig. 4 The encapsulation, growth and release of cells from mCNC hydrogel. Gelation is achieved by heating to 37 °C, and gel dissociation occurs by cooling to room temperature (r.t.). Reproduced from ref. 69 with permission from the American Chemical Society.

Table 2 Some reported thermosensitive polymers and their LCSTs^{70,89}

Thermosensitive polymers	LCST (°C)
Poly(<i>N</i> -isopropylacrylamide) (PNIPAAm)	32
Poly(<i>N,N</i> -diethylacrylamide) (PDEAAm)	33
Poly(methyl vinyl ether) (PMVE)	37
Poly(<i>N</i> -vinylcaprolactam) (PVCL)	34
PEG-based (co)polymers	20–85
Methyl cellulose (MC)	32
Hydroxypropyl cellulose (HPC)	45
Hydroxypropylmethyl cellulose (HPMC)	69
Ethylhydroxyethyl cellulose (EHEC)	Not reported
Hydroxyethyl cellulose (HEC)	Not reported
Poly(asparagine) derivatives	28–78

Generally, thermosensitive polymers are characterized by the presence of different hydrophobic groups, such as methyl, ethyl and propyl. These thermo-responsive polymers include PNIPAAm and its derivatives, poly(*N,N*-diethylacrylamide) (PDEAAm), poly(methyl vinyl ether) (PMVE), poly(*N*-vinylcaprolactam) (PVCL), and PEG-based (co)polymers.^{70,89} Thermo-responsive CNC/PVCL hydrogels were prepared by a frontal polymerization technique.⁷⁰ CNC was found to reinforce PNVCL polymer hydrogels without influence on the LCST which was located at 33–34 °C, a value that is closer to the physiological one than that of PNIPAAm (32 °C). Besides, many natural thermo-responsive polymers, such as methyl cellulose (MC), hydroxypropyl cellulose (HPC), and hydroxypropylmethyl cellulose (HPMC) have also been developed for the control of the thermosensitivity and gelation temperature of cellulose-based hydrogels.^{9,71} The LCSTs of some thermosensitive polymers are listed in Table 2. The LCST can be changed by adjusting the ratio of hydrophilic and hydrophobic groups in the polymer chain. The LCST becomes lower if any hydrophobic group is present in the polymer chain, and a higher LCST will be obtained if any hydrophilic group is present in the polymer.

Other single-responsive cellulose-based hydrogels. Except for pH- and thermo-responsive hydrogels, gas-sensitive cellulose-based hydrogels have also been developed, such as carbon dioxide (CO₂)-switchable hydrogels.^{72,73} Oechsle *et al.* prepared CO₂-switchable hydrogels by adding imidazole to a suspension of CNCs.⁷² By using a typical sulfuric-acid-catalyzed hydrolysis route, the CNCs would have some sulfate half-ester groups on their surface, which give CNCs an overall negative charge and render them dispersible in water due to the electrostatic repulsion. Sparging of CO₂ through the imidazole-containing CNC suspension led to gelation of the CNCs, which was reversible by subsequent sparging with N₂ to form a low-viscosity CNC suspension. The CO₂-triggered gelation was attributed to an increase in ionic strength due to the protonation of imidazole and the formation of bicarbonate in solution. The significant increase in ionic strength resulted in the destabilization of CNCs in the suspension, thus forming a gel. In this way, the gelation of CNC hydrogels can be switched by simply sparging CO₂ or N₂, without the need for any extensive chemical functionalization, or the addition of acids and bases as a trigger. Such CO₂-responsive CNC hydrogels may have potential in

various applications, such as stimuli-responsive adsorbents, flocculants, and tissue engineering.

3.1.2 Multi-responsiveness. Multi-responsive cellulose-based hydrogels are sensitive to more than one environment stimulus, and these stimuli can lead to contraction, swelling, degradation, colour and transparency change, *etc.* Rationally designed stimuli-responsive cellulose-based hydrogels have attracted considerable attention for biomedical applications. For example, pH- and temperature-responsive cellulose-based hydrogel systems have been extensively studied for controlled drug release.^{75,77,78} These dual stimuli-responsive systems can be achieved by using polymers containing various functional groups. Chen *et al.* synthesized dual pH- and temperature-sensitive hybrid hydrogels by incorporating cellulose nanowhiskers and carboxymethyl chitosan into chemically crosslinked PDMAEMA hydrogels through *in situ* radical polymerization.⁷⁵ The pH sensitivity was attributed to the protonation of the tertiary amine groups in PDMAEMA polymer chains, which could be protonated in a low pH medium and are rendered hydrophilic, whereas they became hydrophobic due to deprotonation at high pH. The gels also exhibit thermosensitivity due to the temperature-controlled hydrogen-bond interaction between tertiary amine groups and water. The LCST was found to be 35–40 °C, which is close to the LCST of pure PDMAEMA, suggesting the slight effect of an additive on the thermal responsiveness of gels. Flutamide was used as a model drug to explore the potential of the gels as a drug carrier. A sustained drug release performance was achieved, suggesting the gels could be considered as a candidate for applications in drug delivery.

By changing the copolymerization monomers, various multi-responsive cellulose-based hydrogels can be obtained. pH- and salt-responsive hydrogels were developed by chemical cross-linking of CMC and quaternized cellulose (QC) in NaOH aqueous solution using ECH as a cross-linker.⁷⁴ The strong electrostatic interactions between acidic groups ($-\text{COO}^-$) in CMC and basic groups ($-(\text{CH}_3)_3\text{N}^+$) in QC on the polysaccharide backbone were found to be responsible for the pH responses of the hydrogels in solution. Smart swelling behaviour was achieved in NaCl, CaCl_2 , and FeCl_3 aqueous solutions, while the swelling ratio of the hydrogels decreased with an increase of the salt concentration on the whole.

Shi *et al.* reported a pH and electric field stimuli-responsive release system that was constructed by bacterial cellulose nanofibers and SA hydrogels loaded with ibuprofen.⁷⁹ The drug release rate was controlled based on the variation of pH and electric stimuli. Phenylboronic acid (PBA) and its derivatives can form reversible covalent complexes with polyol compounds such as glucose. In an aqueous medium, PBA compounds are often in equilibrium between undissociated (uncharged) and dissociated (charged) forms. Thanks to these properties, a novel pH- and glucose-responsive composite hydrogel system was obtained from the combination of cellulose and 4-vinyl-PBA.⁸⁰ The self-regulation of insulin release from the gel was achieved by varying the concentrations of glucose solutions. Such an intelligent and biocompatible drug delivery system has great potential in biomedical fields.

Besides, pH and redox responsive cellulose-based hydrogels have also been reported for various biomedical applications.^{81,82} For instance, carboxyethyl cellulose-*g*-dithiodipropionate dihydrazide (CEC-TPH)/dibenzaldehyde-terminated poly(ethylene glycol) (PEG-DA) hydrogels were constructed through dynamic covalent acylhydrazone linkages under 4-amino-DL-phenylalanine (4a-Phe) catalysis.⁸² The prepared CEC-TPH/PEG-DA hydrogels displayed pH/redox dual responsive sol-gel transition behaviours, due to the reversible breaking and reforming of the acylhydrazone and disulfide linkages. Such a dual-responsive hydrogel system was applied to the controlled release of doxorubicin (DOX). More importantly, benefiting from the excellent biocompatibility and the reversible cross-linked networks, the hydrogels were demonstrated to function as suitable 3D culture scaffolds for L929 cells, leading to the encapsulated cells maintaining a high viability and proliferative capacity. Therefore, the CEC-TPH/PEG-DA hydrogels show potential applications in drug delivery and 3D cell culture for tissue engineering.

Reduction sensitive biodegradable cellulose-based hydrogels are also obtained by using di-sulphide cross-linkers, since the di-sulphide bond can be cleaved by various reducing agents, such as dithiothreitol and glutathione tri-peptide (G-glutamyl-cysteinyl-glycine, GSH), due to thiol-disulphide exchange reactions. For instance, pH, temperature and redox responsive copolymer hydrogels were prepared from methacrylated CMC and PNIPAAm by a free radical polymerization technique using *N,N'*-bis(acryloyl)cystamine as a di-sulphide redox sensitive cross-linker.⁸⁵ Lysozyme was used as a model drug and the multi-sensitive release behaviours of drug-loaded hydrogels were demonstrated in different pH (1.2 and 7.4), at different temperatures (25 °C and 37 °C), and in the presence of GSH (20 mM). The release rate was found to be a maximum at lower temperature, lower pH and in the presence of GSH. Since GSH is a common intracellular reducing molecule, the intracellular GSH concentration (~ 10 mM) is significantly higher than that of the extracellular environment, and such a multi-responsive hydrogel may be used as a versatile carrier for intracellular drug delivery.

3.2 Mechanical properties

Mechanical properties including mechanical strength, flexibility, and toughness are well-studied characteristics of cellulose-based hydrogels. In many cases, improved strength was related to higher transparency, and similarly depended on the degree of phase separation.⁴ For example, hydrogels with improved tensile strength were found with smoother surfaces, more homogeneous structure, and smaller pore sizes.⁹⁰ Compared with conventional single network hydrogels that usually fail at tensile stress less than sub-MPa and strain less than 100%, double-network hydrogels and nanocomposite hydrogels often exhibited higher mechanical strength and toughness.^{91,92} Generally, chemical hydrogels exhibit higher strength than physical hydrogels. However, the chemical reactions will lead to permanent junctions in the polymer network; on the contrary, the physical reactions are reversible. In order to achieve enhanced mechanical performance, dual cross-linked hydrogels, such as dual

physically cross-linked hydrogels,⁵² and physical-chemical hybrid cross-linked hydrogels,^{41,93,94} have been developed. In these hydrogel networks, chemical cross-linking points could maintain the elasticity of hydrogels, while the physical cross-linking points promote the viscoelastic and recovery properties.

As shown in Fig. 5, the double-cross-linked (DC) cellulose hydrogels constructed by using sequential chemical and physical cross-linking were demonstrated to be mechanically superior to single-cross-linked cellulose hydrogels, including only physically cross-linked (PC) and only chemically cross-linked (CC) cellulose hydrogels.⁴¹ In this study, the chemical cross-linking domains with irreversible covalent bonds between cellulose chains were obtained by introducing ECH as a cross-linker in a cellulose/LiOH/urea aqueous solution. After the chemical cross-linking reaction, the residual ECH was removed to allow for the formation of physically cross-linked domains by hydrogen bonding and chain entanglements between cellulose chains, as well as crystallite hydrates of cellulose II. The DC hydrogels were foldable and could quickly recover their original shape without any rupture after compression, however, both the PC and CC hydrogels ruptured easily when subjected to bending or large deformations (Fig. 5A–D), demonstrating the excellent flexibility and elasticity of the DC hydrogels. Moreover, the DC hydrogels could be easily formed with a required shape using different molds, such as flower and ball shapes (Fig. 5E and F). A DC hydrogel ball was used as a ping pong ball to play without rupturing in this study (Fig. 5F), which further demonstrated the high recoverability in mechanical properties.

The effects of chemical cross-linking and physical cross-linking on the mechanical properties of DC hydrogels were evaluated under compression and tension models. It was found that the mechanical properties of the DC cellulose hydrogels

were superior to those of both PC and CC hydrogels. The maximum stress at fracture and fracture strain values of DC hydrogels were 4.8 MPa and 74%, respectively. The fracture energy of DC hydrogels reached a maximum value of 0.65 MJ m^{-3} , which was 1150 times greater than that of CC hydrogels, indicating a significant reinforcement in toughness. The highest tensile strength, elongation at break, and fracture energy were 2.7 MPa, 81%, and 0.85 MJ m^{-3} , respectively. The fracture energy of the DC hydrogel was 2.4 times and 500 times greater than those of PC and CC hydrogels, respectively. Moreover, no yielding or necking phenomenon was observed on the DC hydrogels, which exhibited a strain-hardening behaviour driven by an entangled network of cellulose chains and strong hydrogen bonds between the cellulose chains. Hence, the prepared DC cellulose hydrogels with substantial improvement in the mechanical properties and tensile strength have potential in various biomedical applications such as artificial blood vessels and skin, tissue engineering materials, *etc.*

Cellulose NPs, especially CNCs, are commonly used as reinforcing agents for the preparation of composite hydrogels due to their unique properties such as high strength, light weight, biodegradability, and general biocompatibility.^{31,39,52,95–97} For example, Yang and co-workers reported tough and stretchable CNC–PAAm composite hydrogels containing both covalent cross-links and noncovalent hydrogen bonds.³¹ Monotonic stress–strain measurements found that the prepared hydrogels possessed both higher Young's modulus and fracture strength than the pristine PAAm gel, and the composites elongated with increasing content of CNCs from 0.1% to 0.6 v/v%. The modulus of CNC0.6 (prepared with a CNC volume content of 0.6%) was up to 33.8 kPa, which was increased by 6.1-fold compared to that of the native PAAm hydrogel. They proposed that such a high reinforcement might be attributed to the effective interfacial interaction between CNC and PAAm as well as the homogeneous dispersion of CNC. Similarly, Zhang and co-workers fabricated a series of dual physically cross-linked hydrogels based on poly(acrylic acid-*co*-acrylamide) (PAAAm) and quaternized tunicate CNCs.⁵² CNCs acted as both reinforcements and multifunctional crosslinkers in the nanocomposite hydrogels to form loose cross-linking, whereas compact cross-linking was built by ionic coordination between Fe^{3+} and COO^- of PAAAm. The toughness of the dual cross-linked hydrogel was found to be 340 times that of the mono-crosslinked hydrogel, and was 10 times that of the PAAAm hydrogel.

3.3 Self-healing properties

Self-healing hydrogels, also known as self-repair or self-recovery hydrogels, have been developed with inspiration from the biological world where some living tissues, such as skin, bones, and muscle tissues, are capable of autonomously healing when they undergo damage. Self-healing properties can enable hydrogels to intrinsically and automatically heal damage (*i.e.* cut into halves), restoring their original shape/structure and functionality.¹⁶ Self-healing hydrogels with good viscoelasticity, transparency, and biocompatibility have attracted great research interest due to their potential applications in wearable devices,

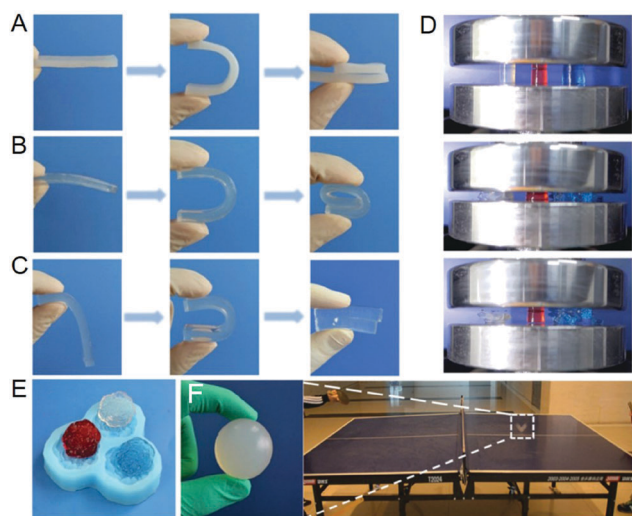


Fig. 5 Photographs of PC (A), DC (B), and CC (C) cellulose hydrogels under bending. (D) The compression behaviour of the PC hydrogel (white), DC hydrogel (red), and CC hydrogel (blue) before (top), during (middle), and immediately after compression (bottom). (E) Photograph of the DC hydrogel flowers. (F) Macroscopic views of a DC hydrogel ball and the hydrogel used as a ping-pong ball. Reproduced from ref. 41 with permission from Wiley.

tissue engineering, biomedicine, artificial skin, and hemostatic materials, *etc.*^{54,98} Hydrogels with both mechanically robust and rapid self-healing properties still remain a challenge, since self-healing hydrogels generally possess either mechanically robust or rapid self-healing properties but not both.⁵⁴ One solution is to design and fabricate self-healing hydrogels by incorporating dynamic covalent and noncovalent bonds into a single hydrogel network. These dynamic covalent bonds, such as Schiff bases,⁹⁹ disulfide bonds,¹⁰⁰ acylhydrazone linkages,^{82,101} and Diels–Alder reactions,¹⁰² can integrate both the stability of covalent bonds and the reversibility of non-covalent bonds.

Yang *et al.* constructed cellulose-based self-healing hydrogels based on CEC–TPH and PEG–DA *via* dynamic covalent acylhydrazone linkages under 4a-Phe catalysis.⁸² The self-healing abilities of the CEC–TPH/PEG–DA hydrogels are shown in Fig. 6. The two stained hydrogel disks were cut in half; after keep the semicircles in intimate contact along the cut line at room temperature for 6 h, without any external intervention, they completely merged into an integral hydrogel disk (Fig. 6A), and the resultant hydrogel was strong enough to withstand a tensile force along the direction vertical to the cut surface without splitting (Fig. 6B-a). The self-healing process was also recorded by optical microscopy (Fig. 6B-b–d), in which the two dye molecules continually spread across the cut surfaces and finally interpenetrated with each other to present a purple colour at the boundary. By the strain compression test, the fracture strength of the self-healed hydrogels was found to be increased with the increasing of healing time, and finally reached a value close to the original one (Fig. 6C), indicating the self-healed

hydrogels could well restore their original mechanical properties. It was also found that the healing efficiency (HE) relied on the healing time and 4a-Phe content, and a high HE of about 96% was achieved at 12 h (Fig. 6D). They attributed these self-healing properties to the acylhydrazone exchange reaction in the hydrogel networks, and the 4a-Phe could accelerate the equilibration kinetics of the acylhydrazone formation and transamination. Thus, when a crack occurred, the CEC–TPH and PEG–DA macromolecules would move toward the damaged interface and the reversible acylhydrazone linkages reformed, resulting in the recovery of the hydrogel networks.

In another case, Shao *et al.* built self-healing cellulose-based hydrogels with toughness and high resilience by furyl-modified CNCs and maleimide end-functionalized PEG *via* a thermally reversible covalent Diels–Alder click reaction.¹⁰² The hydrogels displayed outstanding mechanical properties with a fracture elongation up to 690% and a fracture strength up to 0.3 MPa at a strain of 90%. The self-healing capability of the composite hydrogels assessed by tension tests was found to be 78%. They also developed a tough and self-healing gel by constructing synergistic multiple coordination bonds among TA-coated CNCs, PAA chains, and Al^{3+} ions in a covalent polymer network.³⁰ The incision of the PAA–TA@CNC– Al^{3+} gel was found to self-heal automatically and disappeared almost completely in a short time (within 30 min), and the self-healed gel was strong enough to sustain stable self-supporting and stretching without failure at the interface (Fig. 7A). The fracture stress of the healed gel displayed significant enhancement with increasing healing time (Fig. 7B), and reached an equilibrium state with HE up to 92%

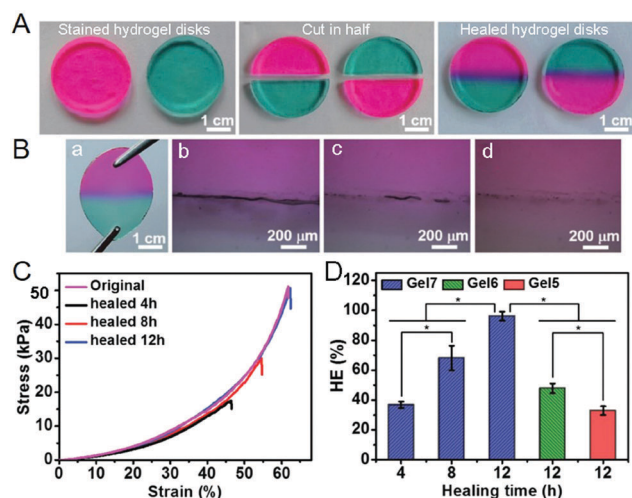


Fig. 6 (A) Photographs of the self-healing process of Gel8; after healing for 6 h at 25 °C, the two semicircle hydrogels were healed into an integral hydrogel disk. (B) Self-healed hydrogel Gel8 under stretching (a), and the optical microscope images of the cut surfaces with healing times of (b) 0 h, (c) 2 h, and (d) 6 h, respectively. (C) Typical compressive stress–strain curves of Gel7 before and after healing for 4, 8, and 12 h. (D) Effects of healing time and 4a-Phe content on the HE of the hydrogels. * indicate $p < 0.05$. (Gel5, Gel6, Gel7, Gel8 refer to the hydrogels prepared with 4a-Phe contents of 0.35, 0.63, 1.05, 2.10 wt%, respectively.) Reproduced from ref. 82 with permission from Wiley.

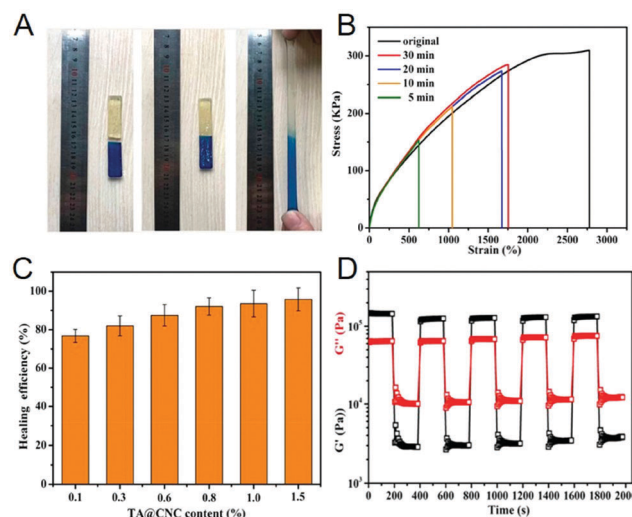


Fig. 7 Self-healing properties of the PAA–TA@CNC– Al^{3+} gel at 25 °C. (A) Methylene blue stained gel is cut into two pieces and recombined. Then the fractured gel can self-heal automatically into a complete gel and sustain stretching without failure after 30 min healing. (B) Typical stress–strain curves of the TA@CNC-0.8 (refers to the gel prepared with 0.8 wt% of TA@CNC against AA) gel before and after healing for different times. (C) HE of gels as a function of TA@CNC content after 30 min healing. (D) G' and G'' of the TA@CNC-0.8 ionic gel in the alternate step strain test (strain = 1 and 100%) at a fixed time interval of 200 s. Reproduced from ref. 30 with permission from the American Chemical Society.

after 30 min healing against the original one. The HE was found to be related to the TA@CNC content, and the increase of TA@CNC content was able to improve the HE of the gel with 30 min healing (Fig. 7C), highlighting the crucial role of dynamic TA@CNC motifs for the reversible rearrangement in the self-healing process. The microscopic self-healing behaviour of the gel was also proved by rheological experiments studying the breakup and reformation of the gel network (Fig. 7D). As expected, by switching the strain from 100% to 1% at a fixed frequency, the gel-like character ($G' > G''$) was recovered instantaneously without any significant decrease in each repeatable cycle of the recovery. Such a sol-gel transition with the complete recovery of the gel network after disruption was attributed the reconstruction of reversible ionic coordination complexation in the gel network system.

4. Applications of cellulose-based hydrogels

The fascinating physicochemical properties of cellulose-based hydrogels make them promising for various biomedical applications. Up to now, the specific applications of cellulose-based hydrogels in the biomedical field can be classified into five parts, including drug delivery, tissue engineering, wound dressing, bioimaging, and wearable epidermal sensors.

4.1 Drug delivery

Smart hydrogels with high water content, soft and rubbery consistency and low interfacial tension with water and biological fluids, are considered as excellent drug delivery systems for various bioactive molecules. Characteristics like the swelling behaviour in an aqueous medium, pH and/or temperature sensitivity, and zero-order kinetics play important roles in the development of hydrogel-based drug delivery systems. The gel solution can be loaded with bioactive compounds, such as drugs,^{19,42,83} proteins,⁷⁸ and cell factors,^{20,103} and the porosity and hydrophilicity of the hydrogel can regulate the loading and release of drugs in a controlled manner. In such a controlled drug release behaviour, the hydrogel carriers can protect the structural integrity and bioactivity of the drugs from the environment, and their biodegradability can be modified according to the route of administration. For instance, a hydrogel system for oral protein delivery should be biocompatible and pH responsive to protect proteins from the harsh stomach environment and prolong the hydrogel residence time in the intestines, as well as capable of decreasing permeation barriers in the intestine.⁴⁸ Hydrogels can be loaded with drugs by premixing the drug in the reaction system before preparing hydrogels,¹⁰³ or by physical adsorption (e.g. a swelling diffusion method) that reswells dried hydrogels in drug solutions,^{48,65,104} or by conjugating the drug to the cellulose backbone to prepare prodrug hydrogels.¹⁰⁵ The interactions between the hydrogel and drug include covalent bonding and noncovalent bonding (such as hydrogen bonding, hydrophobic interactions, π - π stacking, and electrostatic interactions).^{42,62,105}

The pH-responsive bacterial cellulose (BC)-g-PAA hydrogels prepared by EB irradiation were explored as an oral delivery system for proteins by using bovine serum albumin (BSA) as a model protein drug.⁴⁸ BSA was loaded into the hydrogels *via* a swelling diffusion method, the entrapment efficiency of BSA was 40–55%, and hydrogels with greater swelling exhibited greater EE. Drug release was carried out in simulated gastroenteric fluids, in which the BSA-loaded hydrogels were first released in simulated gastric fluid (SGF, pH 1.2) for 2 h and then transferred to simulated intestinal fluid (SIF, pH 6.8) until maximum release. It was found that less than 10% of BSA was released in SGF due to the lower swelling of hydrogels at acidic pH, demonstrating the potential of the hydrogels to protect BSA from the acidic environment of the stomach. When the same hydrogels were moved to SIF, a rapid increase in BSA release was observed due to higher swelling of the hydrogels at higher pH, and a maximum release was reached of about 90%. Subsequent conformational stability analyses indicated that the structural integrity and bioactivity of released BSA were maintained and preserved by the hydrogels. Moreover, increased BSA penetration across intestinal mucosa tissue was observed in an *ex vivo* penetration experiment, and the acute oral toxicity in Wistar rats demonstrated the safety of these hydrogels by oral administration. Therefore, the prepared BC-g-PAA hydrogels have potential for further development as oral protein delivery carriers.

As an amphiphilic anticancer drug with high efficiency and a broad spectrum of activity against diverse cancer types, DOX suffers from poor solubility in water and physiological media (solubility of hydrochloride salt <2%). To solve this problem, Capanema and co-workers conjugated DOX to the CMC backbone by covalent amide linkages between the $-\text{COO}^-$ groups in CMC and $-\text{NH}_2$ groups in DOX using 1-ethyl-3-[3-dimethylaminopropyl]carbodiimide hydrochloride (EDC) as a “zero-length” coupling agent in 2-(*N*-morpholino)ethanesulfonic acid (MES) buffer.¹⁰⁵ The obtained CMC-DOX was water soluble, and then, CMC-DOX hydrogels were produced by using eco-friendly citric acid as a crosslinking agent; the drug loading efficiency was found to be more than 98%. Since the degree of substitution (DS) has great influence on the CMC properties, such as the water solubility, pH sensitivity, chemical reactivity and stability, rheology and biodegradability, CMC with different DS (1.22 and 0.77) was used in this study. *In vitro* release experiments showed that the drug release kinetics were influenced by the DS of CMC, and the maximum release of DOX from prodrug hydrogels was $25 \pm 2\%$. The cytotoxicity of the prodrug hydrogels also related to the DS; the time of 50% ($t_{50\%}$) cell viability response was 3.3-fold (CMC1.22-DOX) and 4.5-fold (CMC0.77-DOX) delayed for A375 melanoma cells, as well as 5.0-fold (CMC1.22-DOX) and 6.9-fold (CMC0.77-DOX) delayed for HEK 293T cells, compared with free DOX (no difference between $t_{50\%}$ for cancer cells and normal cells). This delayed effect of toxicity was attributed to the amide bonds between CMC and DOX that require cleavage inside the cell for drug release with the specific biochemical behaviour of different cell types, thus the

association of CMC with toxic DOX can effectively minimize the adverse side effects.

4.2 Tissue engineering

The soft tissues of the human body, such as tendons, ligaments and cartilage, are excellent “soft & wet” materials with superior mechanical performance and 30–80% water.⁹¹ Similar to biological tissues, polymer hydrogels that consist of 3D polymer networks and a large amount of water, also have a “soft & wet” nature, endowing them with great potential for application in tissue engineering.^{91,106} However, most polymeric hydrogels cannot meet the requirements (including the mechanical strength, toughness, extensibility, and recoverability) of these load-bearing soft tissues, and are thus limited to usage where mechanical properties are undemanding such as drug delivery and water absorbents.^{49,103,107} Recently, some successful strategies have been constructed to prepare hydrogels with enhanced mechanical performance and other attractive properties, such as double network hydrogels^{41,52} and nanocomposite hydrogels,^{22,108,109} which are promising for tissue engineering applications.

CNCs, which are typically isolated from cellulose, have been widely used as fillers to reinforce polymeric hydrogels due to their favourable mechanical properties and intrinsic biocompatibility.^{31,59,95–97} The unique aspect ratio and plenty of active –OH groups on their surface make CNCs a remarkable precursor to prepare a solid stabiliser.¹¹⁰ Yang *et al.* reported injectable hydrogels based on adipic acid dihydrazide-modified CMC and aldehyde-modified dextran reinforced with CNCs and aldehyde-functionalized CNCs (CHO-CNCs).⁹⁷ Gelation occurred within seconds as the hydrogel components were extruded from a double-barrel syringe, and the CNCs were observed to be evenly distributed throughout the composites. Swelling tests showed that all the CNC reinforced hydrogels could maintain their structure for more than 60 days both in water and 10 mM PBS, suggesting the possibility for longer-term applications. The CHO-CNCs could act as both a filler and a chemical cross-linker, making the CHO-CNC-reinforced hydrogels more elastic, more stable, and capable of facilitating higher NP loadings without sacrificing mechanical strength compared to hydrogels with unmodified CNCs. Cytotoxicity tests found that both the starting components and the hydrogels showed good cytocompatibility against NIH 3T3 fibroblast cells. Combined with their syringeability, the prepared CNC-reinforced injectable hydrogels could fill in irregular cavities and shapes without the need for pre-shape forming processing, thus they were promising for bone tissue engineering.

The introduction of inorganic NPs, such as LAPONITE[®],²² hydroxyapatite (HAp),^{108,109} and titanium dioxide¹¹¹ can also reinforce cellulose-based hydrogels for bone tissue engineering applications. Boyer and colleagues developed a LAPONITE[®] NP reinforced silated HPMC (Si-HPMC) hydrogel, in which the LAPONITE[®] NPs were self-setting within the gel structure of Si-HPMC, resulting in a hybrid interpenetrating network (IPN).²² This IPN structure increased the mechanical properties of the hydrogel without interfering with the O₂ diffusion and

cell viability after gelification. The ability of the hybrid scaffold containing the Si-HPMC/LAPONITE[®] hydrogel and chondrogenic cells to form cartilaginous tissue *in vivo* was investigated during 6 week implantation in subcutaneous pockets of nude mice. Histological analysis of the composite constructs revealed the formation of cartilage-like tissue with an extracellular matrix containing glycosaminoglycans and collagens. These results indicated the prepared Si-HPMC/LAPONITE[®] hydrogels have great potential for cartilage defect repair. Besides, Tohamy *et al.* constructed composite scaffolds of SA/HEC/HAP *via* a lyophilization technique and further cross-linking in the presence of Ca²⁺ ions.¹⁰⁹ It was found that the higher HAP concentration (40 wt%) could effectively enhance the mechanical properties (23.9 MPa), bioactivity and protein adsorption. Cell experiments confirmed the non-toxicity and good proliferation capability of the SA/HEC/HAP scaffold, which may be attributed to the biocompatibility, bioactivity, perfect cell adhesiveness, and good mechanical strength of HAP.

The morphology, inner pore structure, and uniformity of hydrogels play an important role in determining their biomedical applications. 3D printing technology provides a unique opportunity to develop biomedical scaffolds with uniform or gradient porosity required for tissue repair and regeneration using biocompatible cellulose-based nanocomposites.^{59,112} By 3D printing technology, Sultan *et al.* fabricated CNC-based hydrogel scaffolds with uniform and/or gradient porosity from a hydrogel ink of SA and gelatin reinforced with CNCs.⁵⁹ The CNCs provided favourable rheological properties required for 3D printing. The printed scaffolds were crosslinked sequentially *via* covalent and ionic reactions, resulting in dimensionally stable hydrogel scaffolds with pore sizes of 80–2125 μm and nano-scaled pore wall roughness, which were favourable for cell interaction. Moreover, they have constructed a new nozzle movement, nozzle movement-B, for 3D printing hydrogel scaffolds with uniform and gradient pore structures (Fig. 8A). In this way, the walls of each pore were printed once, and a higher Z-axis (up to 20 mm) was achieved with good printing resolution.

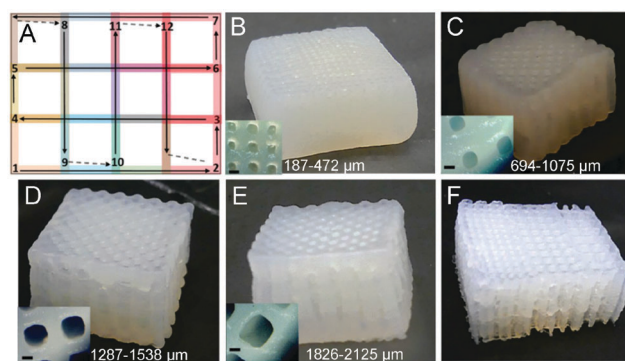


Fig. 8 (A) Schematic representation of nozzle movement-B used for 3D printing hydrogel scaffolds. The dashed arrow represents the nozzle jump. Images and pore size ranges of the 3D printing hydrogel scaffolds (B) 0.5 mm pore size, (C) 1 mm pore size, (D) 1.5 mm pore size, (E) 2 mm pore size, and (F) gradient pore size. Adapted from ref. 59 with permission from the Royal Society of Chemistry.

With this mode, four scaffolds with uniform pore sizes and one gradient porous scaffold were successfully printed (Fig. 8B–F). The orientation of the nanostructures and nanoscale pore wall roughness in hydrogel scaffolds were considered to be suitable for cell interactions and guided cell growth during tissue regeneration.

4.3 Wound dressing

Skin is the largest organ in the human body, which can protect the body from harmful environments and keep the balance of water and electrolytes.^{23,35} The human body is capable of restoring skin integrity and homeostasis after injury, while the healing process can be compromised by many factors, such as aging, disease, nutritional status and local factors (e.g. infection, tissue ischemia, and mechanical pressure), leading to scar formation or chronic wounds.²³ Serious tissue damage can result from infection caused by microorganisms, such as *Staphylococcus aureus* (*S. aureus*), *Escherichia coli* (*E. coli*), *Pseudomonas aeruginosa* (*P. aeruginosa*), and rapid wound treatment has therefore received much attention.^{24,25,113–115} Wound dressings are widely used to support the wound bed and protect it from contamination and moisture loss, thus the wound dressing should have certain abilities of antibacterial activity, water vapor permeability, oxygen permeability, and mechanical strength. Cellulose-based hydrogels possess 3D hydrophilic networks and can swell by tens or even hundreds of times in volume after absorbing a large amount of water, thus they can provide a moist environment that is required for wound healing. After loading antimicrobial agents (such as Ag, AgCl, and ZnO)^{25,113,115} or antibiotics (such as tetracycline hydrochloride),¹¹⁶ these hydrogels can absorb tissue exudates and kill bacteria in wounds. In consideration of the essential biocompatibility, biodegradability, and environmentally-friendly properties of cellulose, cellulose-based hydrogels are promising wound dressing materials.¹¹⁷

Anjum *et al.* developed a composite material containing nanosilver nanohydrogels (nSnH) along with *Aloe vera* and curcumin that promote antimicrobial activity, wound healing and infection control.¹¹³ A PVA/polyethylene oxide/CMC matrix was used as a gel system to blend with nSnH, *Aloe vera*, and curcumin, and was then coated onto hydrolysed polyester fabric to develop antimicrobial dressings. The cumulative release of Ag⁺ from the dressing was found to be about 42% of the total loading after 48 h. The antimicrobial activity of the dressings was demonstrated against both *S. aureus* and *E. coli*; all of the dressings showed bacteria colony reduction above 70%, and the Gel/nSnH/*Aloe* was found to lead to complete (100%) reduction of *S. aureus*. *In vivo* wound healing was carried out on full-thickness skin wounds created on Swiss albino mice. After 16 days, the Gel/nSnH/*Aloe* treated wounds were healed 100% with minimum scarring. By histological studies, Gel/nSnH/*Aloe* treated wound tissues were found to exhibit a thinner epidermal layer and organized collagen deposition without inflammation of healed tissues. These results indicated that nSnH and *Aloe vera* based dressing materials could be promising candidates for wound dressings.

In another case, a CMC hydrogel embedded with Ag/Ag@AgCl/ZnO nanostructures was found to have excellent antibacterial activity and wound healing acceleration.²⁵ The CMC hydrogel was prepared by using ECH as a cross linker, and then the Ag/Ag@AgCl nanostructures were assembled in the hydrogel *via* UV light chemical reduction followed by incorporation of ZnO nanostructures by NaOH precipitation. The controllable and sustained release of Ag⁺ and Zn²⁺ ions over a period of 21 days was attributed to the reversible swelling–shrinking transition of the hydrogel triggered by changing pH in the environment. About 90% Zn²⁺ ion release was observed in the acidic environment after 3 days, whereas only 10% Zn²⁺ ion release occurred in the neutral environment after 21 days. The Ag/Ag@AgCl nanostructures were found to effectively enhance the photocatalytic activity and antibacterial activity of ZnO due to the enhancement of reactive oxygen species (ROS) by visible light. The bacteria killing rates for the hydrogel systems were up to 95.95% of *E. coli* and 98.49% of *S. aureus* within 20 min upon exposure to simulated visible light. Such rapid sterilization plays a crucial role in wound healing.

In vivo wound healing abilities of the prepared hydrogels including the pure CMC hydrogel (the control, H1), and the CMC hydrogel containing Ag/Ag@AgCl (H2), Ag/Ag@AgCl/ZnO (H4), or pure ZnO (H6) were evaluated using animal models (Fig. 9). As shown in Fig. 9A, all the groups showed severe bacterial infection with ichor after 2 days of treatment due to the *S. aureus* formed in wound exudate, which was demonstrated by the appearance of many neutrophils through hematoxylin and eosin staining of the midportion of repaired tissues. The H4 group was found to have better antibacterial effects than the other groups after 4-days of treatment, and dermal fibroblasts and newborn blood vessels were found stemming from Zn²⁺ ion release, involved in the function of fibroblasts and production of more fibroblasts. After 8-days of treatment, few viable colony units were formed for the H4 group, and all the hybrid hydrogels showed a smaller trauma size than the H1 group. The H4 and H6 groups healed well after 14 days, while infected necrotic foci (cyan rectangles) appeared in H1 and H2 groups suggesting the wounds did not heal. The number of white blood cells (WBCs) and neutrophils from all the treatment groups showed obviously higher levels than the H1 group at 4-, 8-, and 14-days of treatment (Fig. 9B and C). It was believed that the released Ag⁺ and Zn²⁺ ions could promote the immune functions and produce a large number of WBCs and neutrophils (2–4 times more than H1) leading to synergistic antibacterial activity and accelerated wound healing.

Besides loading antimicrobial agents, cellulose-based hydrogels containing cells (such as keratinocytes and fibroblasts) have also been used to promote wound healing.³⁵ BC/AA hydrogels containing human epidermal keratinocytes (EK) and dermal fibroblasts (DF) were used as a wound dressing for the treatment of a partial-thickness burn wound.³⁵ *In vitro* studies showed that the prepared BC/AA hydrogels exhibited fast cell attachment, in which more than 80% of EK and DF adhered onto the hydrogel within 4 h, and they could maintain cell viability, limit cell migration and allow cell transfer. *In vivo* studies found that the

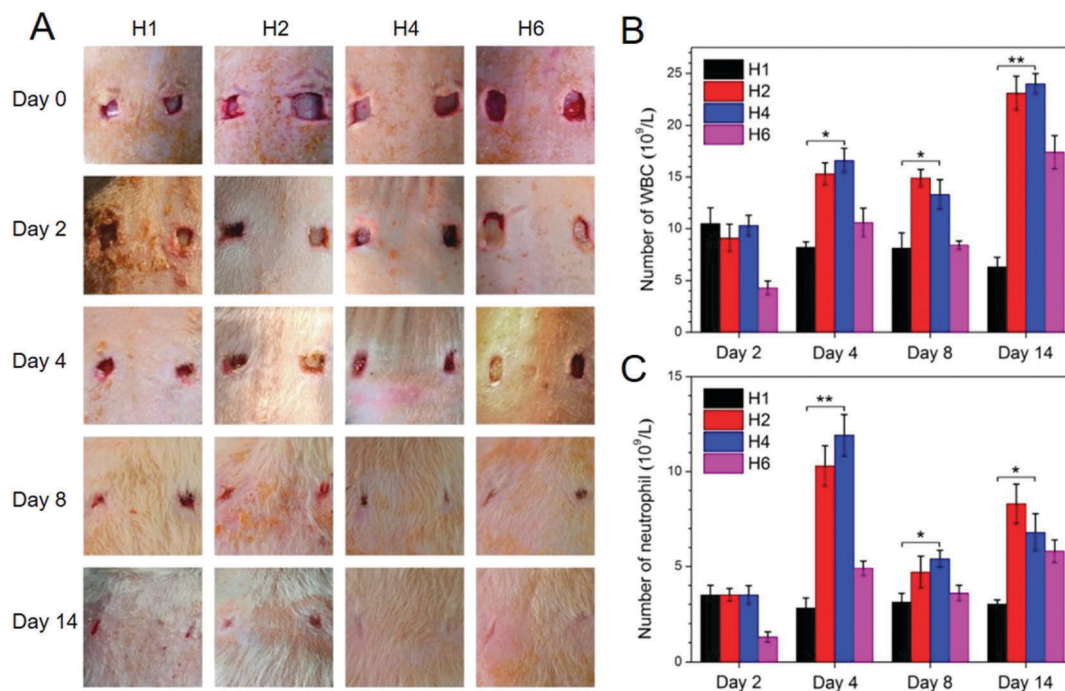


Fig. 9 (A) *In vivo* study on the effects of treatment of *S. aureus*-induced wound infections by hydrogels and the corresponding wound photographs of the rats at days 0, 2, 4, 8, and 14. *In vivo* routine analysis of the numbers of WBC (B) and neutrophils (C) in the whole blood extracted from the rats after treating with hydrogels for 2, 4, 8, and 14 days. H1: CMC hydrogel; H2: CMC–Ag/Ag@AgCl hydrogel; H4: CMC–Ag/Ag@AgCl/ZnO hydrogel; H6: CMC–ZnO hydrogel. The experiment was performed in triplicate and independently ($n = 3$), and data are mean $\pm$ SD. Reproduced from ref. 25 with permission from the American Chemical Society.

wound closure was about 100% for the hydrogel containing cells at 13 days treatment, while the untreated controls and the wounds treated by hydrogel alone did not heal. The wound healing acceleration also proved by histological analysis, immunohistochemistry staining, and Masson's trichrome staining, suggested the potential application of the BC/AA hydrogel as a cell carrier and wound dressing to promote full thickness wound healing.

4.4 Bioimaging

Bioimaging, such as fluorescence imaging (FLI), magnetic resonance imaging, ultrasound imaging, photoacoustic imaging, and computed tomography scanning, has been recognized as an important technology in the life science and medical fields.^{118–120} Fluorescent hydrogels with some unique properties such as injectability, fluorescence and safety, possess great potential in bioimaging applications.^{121,122} Cellulose is a natural renewable polysaccharide on the earth, which has been widely used in various biomedical applications due to its intrinsic biocompatibility, biodegradability, chemical stability, and excellent mechanical strength.^{123–125} By embedding fluorescent materials in cellulose matrices, fluorescent cellulose-based hydrogels can be obtained. These fluorescent materials include CdSe/ZnS quantum dots (QDs),²⁶ rare-earth doped phosphor (SrAl₂O₄: Eu²⁺, Dy³⁺),²⁷ carbon dots (CDs),¹²⁶ and S,N-codoped graphene QDs,¹²⁷ etc. For instance, fluorescent cellulose–QD hydrogels were obtained by embedding CdSe/ZnS QDs in cellulose matrices in a NaOH/urea aqueous system *via* a mild chemical

cross-linking process. The QDs were embedded firmly in cellulose matrices due to the strong interactions (electrostatic attraction and hydrophobic interactions) between the QDs and cellulose after hydrolysis of the QD ligands.²⁶ As shown in Fig. 10A, the obtained cellulose–QD hydrogels were found to exhibit a transparent appearance under visible light, while under a UV lamp, single-color emission (yellowish-green) was obviously observed with very low QD content (0.06 wt%), indicating a high efficiency for the fluorescence emission. QDs with sizes in the range of 2.8 to 3.6 nm (emitting different colors from green to red) were used to prepare cellulose–QD hydrogels. Under UV light, the hydrogels exhibited homogeneous and strong fluorescence from green to red, depending on the size of the QDs (Fig. 10B), suggesting the optical properties of QDs could be well maintained in the hydrogels. The compressive modulus of these hydrogels was found to be about 45–49 kPa, and combined with their good transparency, the prepared fluorescent cellulose–QD hydrogels are promising for applications in the fields of fluoro-immunoassay and biological labelling.

Similarly, a fluorescent cellulose–phosphor (PP) hydrogel was constructed from a cellulose solution containing rare-earth doped PP in a LiOH/urea aqueous system using ECH as a crosslinker.²⁷ The PP particles were found to be tightly embedded in the macroporous cellulose matrix, which could not only supply cavities for PP immobilization through intermolecular hydrogen bonding interactions, but also supply the pore wall as a shell to protect the structure and properties of PP. In this way, the dispersion of PP in aqueous solvent was

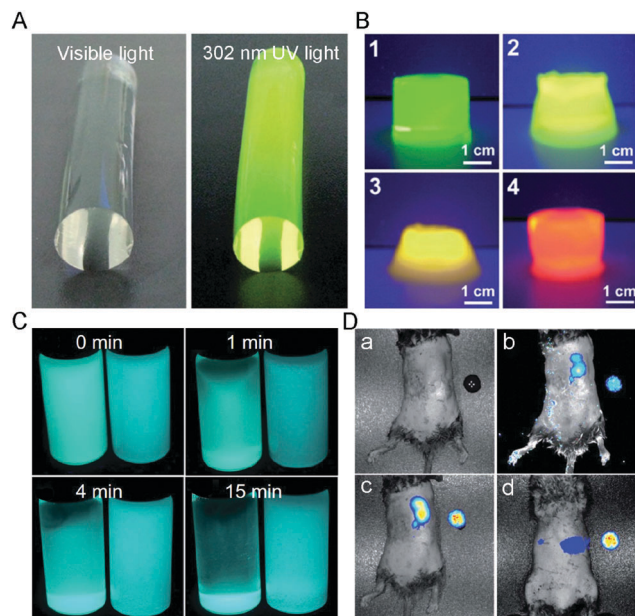


Fig. 10 Photographs of (A) a cellulose-QD hydrogel with 0.06 wt% QDs (3.0 nm) under visible light and a UV lamp, respectively; (B) cellulose-QD hydrogels under a 302 nm UV lamp, the average diameters of the QD nanocrystals are 2.8 nm (green), 3.0 nm (yellowish-green), 3.2 nm (yellow) and 3.6 nm (red). Reproduced from ref. 26 with permission from the Royal Society of Chemistry. (C) 0.25% aqueous solution of PP (left) and a 50% aqueous solution of smashed CPH5 (right) under UV light; (D) a living 129 mouse after injection of CPH5 under the skin (a–c) and in the stomach (d): images in bright-field (a); merged images of the bright-field and the threshold false colour fluorescence image under 455 nm excitation light (b); merged images of the brightfield and the threshold false colour of autofluorescence without excitation light under the skin (c), and in the stomach (d). Reproduced from ref. 27 with permission from Royal Society of Chemistry.

improved significantly. Compared with free PP, the cellulose/PP hybrid (CPH) hydrogels emitted relatively strong green fluorescence under a UV lamp, as well as high brightness and long-lasting afterglow (Fig. 10C). CPH with a strong afterglow could be detected both under the skin and in the stomach with and without excitation light, which was demonstrated by injecting CPH with 0.5 wt% PP content (CPH5) into a mouse (Fig. 10D), showing promising prospects as a candidate for bioimaging.

Fingerprint FLI is one of the most prominent technologies in the field of forensic medicine, but it seldom considers the security protection of detection information. Recently, Hai *et al.* developed a $\text{ClO}^-/\text{SCN}^-$ -stimulated reversible-response lanthanide luminescent Tb(III)-CMC complex hydrogel, which consisted of 2,6-dimethylpyridin-4-amine functionalized with a CMC-binding aptamer and a terbium(III) 2-(2-aminobenzamido)benzoic acid complex modified with CMC, for the selective detection, protection, and storage of fingerprint information.²⁸ As shown in Fig. 11, upon irradiation with 270 nm UV light, the imaging information of the fingerprint could be quenched and recovered by $\text{ClO}^-/\text{SCN}^-$ regulation, respectively, resulting in reversible on/off conversion of the luminescence signals for the encryption and decryption of multiple levels of information.

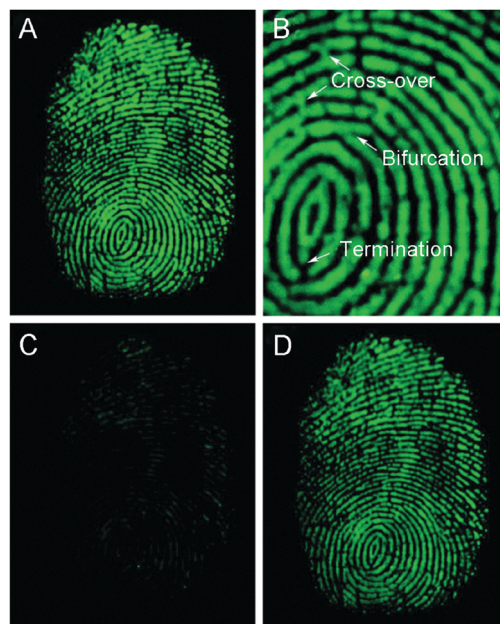


Fig. 11 (A) Luminescence image of a fingerprint on a microscope slide, and (B) corresponding magnified image. (C and D) Luminescence images of the fingerprint in response to ClO^- (C) and SCN^- (D). Reproduced from ref. 28 with permission from Wiley.

The results indicate that fingerprint information detection, encryption, and decryption is possible by the use of the Tb(III)-CMC complex as an invisible security probe, ClO^- as an encryption reagent (Fig. 11C), and SCN^- as a decryption reagent (Fig. 11D). Compared with traditional fluorescent probes, the Tb(III)-CMC complex hydrogel can ensure the confidentiality of fingerprint information, and thus may open up the possibility of using tunable luminescent hydrogels for fingerprint information detection, security protection, and storage.

4.5 Wearable sensors

The development of smart wearable devices has become a research hotspot due to their potential applications in health monitoring. Self-healing wearable devices can restore their structure and functionality after damage, endowing them with enhanced durability, reliability, and safety.¹⁶ As a typical soft and flexible material, self-healing hydrogels have attracted significant attention for the development of self-healing wearable devices (pressure sensors) for human-motion detection, due to their good viscoelasticity, conductivity, and biocompatibility.^{12,17,128} Self-healing hydrogels should simultaneously meet the required mechanical toughness and excellent mobility properties, which are two contradictory characteristics that are derived from their inherent compromise between dynamic cross-links for healing and steady cross-links for mechanical strength. The great challenge is how to integrate excellent self-healing and mechanical properties into a single conductive hydrogel.⁵⁴ CNCs have been commonly used as fillers to reinforce hydrogels due to their unique properties such as excellent mechanical properties, biodegradability, biocompatibility, and modifiability.^{31,95,96} Recently, CNCs have been successfully

used to create integrated conductive hydrogels with both excellent self-healing and mechanical performance.^{29,30}

Inspired by biological soft tissues with hierarchical network structures, Liu and co-workers developed a conductive, elastic, self-healing, and strain-sensitive functional network hydrogel (F-hydrogel) from a “soft” homogeneous polymer network *via* covalent cross-linking of PVA and PVP, in which a “hard” Fe³⁺ cross-linked CNC network with dynamic CNCs–Fe³⁺ coordination bonds was interconnected as reinforcing domains.²⁹ Under stress, the dynamic CNCs–Fe³⁺ coordination bonds acted as sacrificial bonds to efficiently dissipate energy, while the PVA–PVP network led to smooth stress-transfer, which enabled the F-hydrogels to achieve unusual mechanical properties and good self-recovery properties. The F-hydrogels with “soft and hard” hierarchical network structure exhibited excellent mechanical strength (2.1 MPa tensile stress), robust toughness ($\sim 9.0 \text{ MJ m}^{-3}$), and reliable stretchability (830% elongation), making them able to withstand high-level deformations of compression and knotting without any observable damage. The F-hydrogels also showed autonomous self-healing capability in only 5 min without any stimuli or healing agents, and the sensing performance could be well maintained before and after restoration. In addition, the F-hydrogels displayed ultrasensitive, stable and repeatable resistance variations upon mechanical deformations, and could be

applied tightly on the skin surface, which were promising in application of wearable devices.

Based on these characteristics of the F-hydrogels, a wearable soft strain sensor was assembled to monitor finger joint motions, breathing, and even a slight blood pulse (Fig. 12). It was found that when the bending degree of the finger increased from 0° to 120° during a step-by-step bend-holding process, the relative resistance of the F-hydrogel sensor rapidly raised to corresponding values (Fig. 12A), and the F-hydrogel sensor exhibited a repeatable response at a bending degree of 90° and a working frequency of 0.3 Hz (Fig. 12Aa). When the F-hydrogel sensor was attached onto the surface skin of the lower left rib cage of a volunteer, breathing modes (such as regular breathing, rapid deep breathing, and breathing hold) could be monitored and easily distinguished from the relative resistance changes of the F-hydrogel sensor (Fig. 12B). When the F-hydrogel sensor was put on the wrist of a volunteer (Fig. 12C), the blood pulse changes before and after exercise were fully displayed on the relative resistance curve (Fig. 12D), which verified the high strain sensitivity and ultrafast response of the hydrogel sensor, making it an ideal candidate for human health monitoring.

Similarly, Shao *et al.* obtained a tough, self-healing, and self-adhesive gel by constructing synergistic multiple coordination bonds among TA-coated CNCs, PAA chains, and Al³⁺ ions in a

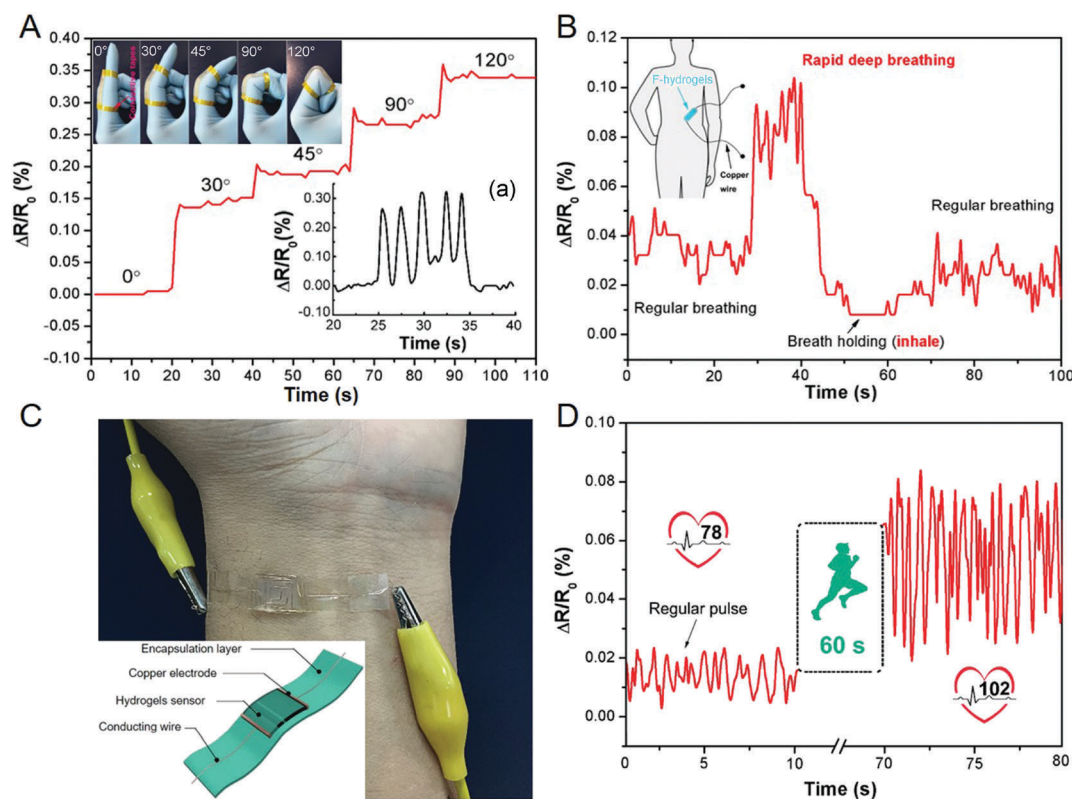


Fig. 12 (A) The relative resistance changes vs. time when bending–unbending the index finger at different angles (inset: photographs of bending the index finger at different angles, the F-hydrogel sensor was fixed on the joints), and (a) a bending cycle of the index finger at 90°. (B) The relative resistance changes vs. time when monitoring human breaths in different breathing modes (the F-hydrogel sensor was attached onto the surface skin of the lower left rib cage). (C) Photograph of the F-hydrogel sensor as a wearable pulse monitor worn by a volunteer (inset: 3D model of the F-hydrogel strain sensor). (D) The relative resistance changes vs. time when monitoring the pulse before and after exercise (a volunteer ran two lengths from the 1st floor to the 4th floor). Reproduced from ref. 29 with permission from the American Chemical Society.

covalent polymer network.³⁰ The incorporated TA@CNC acted as a dynamic connected bridge in the hierarchically porous network mediated by multiple reversible coordination bonds among TA, CNC, and Al^{3+} ions, endowing the PAA-TA@CNCs- Al^{3+} gels with superior mechanical properties, such as ultrastretchability (fracture strain of 2952%), high compression performance (95% strain without fracture), and toughness (5.60 MJ m^{-3}). Also, the dynamic features of multiple reversible coordination bonds endowed the gels with excellent recoverability and electrical self-healing properties. The incision of the gels self-healed automatically and disappeared almost completely within 30 min, and the fracture stress was found to reach an equilibrium state with healing efficiency of 92% after 30 min healing against the original one. Besides, due to the presence of catechol groups from incorporated TA, these gels display durable and repeatable adhesiveness to various substrates (such as human skin tissue, rubber, hogskin, aluminum, glass, and polytetrafluoroethylene) without inflammatory response and residual, which could be employed as flexible strain sensors adhered directly on human skin.

Owing to their excellent self-adhesiveness, high strain sensitivity, remarkable electrical stability, and fast self-healing capability, large motions such as joint bending and stretching of diverse human motions (including the opisthenar, elbow, and shoulder joint bending in a standard shooting action) could be detected by a wearable strain sensor assembled from the gel and directly attached on human skin. Moreover, subtle motions including the pulse and breaths that are beneficial for real-time monitoring of personal healthcare during sports training, could also be accurately detected and promptly recognized by the gel strain sensors (Fig. 13A and B). Importantly, by designing a real-time human-motion monitoring system, the resistance changes of the gel sensors during the motion can be captured and then converted into digital signals by an

analyser (Fig. 13C).¹²⁹ Then, the signals would be transmitted wirelessly to a smart phone, which enables us to analyse the data on the user interface of a smart phone by programmable wireless transmission. Such gels with high strain sensitivity are ideal candidates to assemble stretchable and wearable strain sensors for human activity monitoring and personal healthcare diagnosis. This work also provided a positive exploration to fabricate biocompatible cellulose-based conductive hydrogels for potential applications in wearable electronic sensors, healthcare monitoring, and even soft intelligent robots.

4.6 Other biomedical applications

Besides the aforementioned biomedical applications, many new bioapplications of cellulose-based hydrogels have been explored in recent years, such as enzyme immobilization,⁵¹ blood purification,^{130,131} neural engineering,¹³² and so on.¹³³ For instance, enhanced pH, thermal and storage stability of papain were obtained by immobilization of it on hydrogel composites (pineapple peel CMC/PVA/mesoporous silica SBA-15).⁵¹ It was found that the immobilized papain could retain 56% and 79% of the initial activity after incubation at 80°C for 2 h and storage at 4°C for 10 days, respectively. However, the free papain only retained 16% and 27% of its initial activity at the same conditions. In another case, polymyxin B (PMB) was immobilized on cross-linked cellulose microspheres (CL-CMs) to synthesize an endotoxin adsorbent.¹³⁰ The immobilized PMB CL-CMs showed good adsorption capability for endotoxins. The dynamic adsorption efficiency of endotoxins was 72.3% at a plasma perfusion rate of 300 mL h^{-1} with an endotoxin concentration of 4 EU mL^{-1} , while the variation of plasma protein before and after adsorption was only 8.9%, suggesting the great potential application of the prepared adsorbents in clinical blood purification.

In a word, thanks to their fascinating properties and inherent biocompatibility, cellulose-based hydrogels have been widely studied in biomedical applications. With the development of science and technology, we believe that more bioapplications of cellulose-based hydrogels will be explored.

5. Conclusions and future perspectives

Cellulose-based hydrogels possess attractive and tuneable properties, which make them good candidates for various biomedical applications. Through a selection of preferred preparation methods, cellulose-based hydrogels with desirable properties (such as stimuli-responsive properties, mechanical performances, and self-healing properties) can be obtained and applied for special biomedical applications including drug delivery, tissue engineering, wound dressing, bioimaging, and wearable sensors, *etc.* Although hydrogels based on cellulose and its derivatives have been widely investigated, there are still many challenges that need to be overcome before commercialization.

(1) The long-term biosafety of cellulose-based hydrogel drug delivery systems has not been systematically evaluated. Although the natural cellulose produced by plants or bacteria is intrinsically

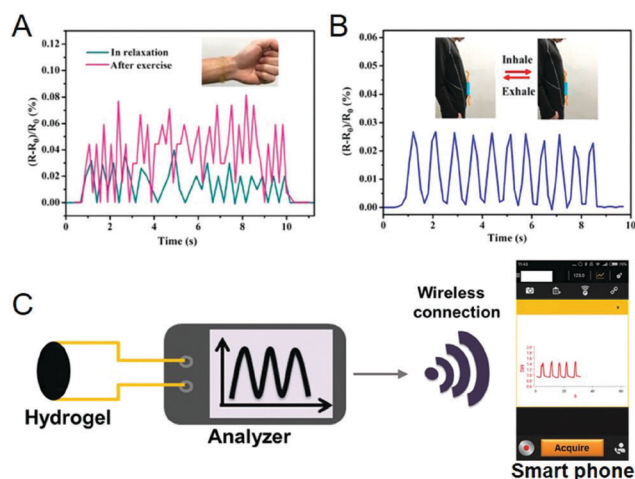


Fig. 13 (A) Real-time resistance change signal of the gel strain sensor adhered onto the radial artery of the wrist to monitor the pulse under relaxation and exercise conditions. (B) Detection of tiny movement caused by breathing. The inset shows the gel sensor adhered on the chest. Reproduced from ref. 30 with permission from the American Chemical Society. (C) Real-time bodily motion monitoring system using a hydrogel strain sensor and smart phone. Reproduced from ref. 129 with permission from Wiley.

biocompatible and non-toxic, and several studies have demonstrated that currently available hydrogel drug delivery systems based on cellulose and its derivatives are biocompatible, most of these studies were based on cell experiments or short-term histopathological assay.^{19,33,105} Systematic investigations on the biosafety of cellulose-based hydrogel drug delivery systems are critical for further biomedical applications. The long-term effects of these drug delivery systems, including long-term toxicity, biocompatibility, immunogenicity, pharmacokinetics, and pharmacodynamics should be confirmed in small animal models (*e.g.* mice, rats) to larger animal models (*e.g.* monkeys) before clinical translation. In order to ensure the biosafety of the hydrogel drug delivery systems, more attention should be paid to the preparation methods, such as the development of green synthesis methods by using green solvents, no or non-toxic cross-linkers, and so on.

(2) Hemostasis, inflammation, proliferation, and remodeling are the four unique yet overlapping stages for the biological wound healing process, while the recently developed cellulose-based hydrogels could only contribute their functions in one or two stages, thus the exploration of multifunctional hydrogel dressings that can work on more stages are needed. For example, by loading and controllably releasing vascular endothelial growth factor from a hydrogel dressing, the processes of cell proliferation and tissue remodelling in the wound bed were significantly improved.⁵ In addition, in order to expand the applications of wound dressings, cellulose-based hydrogels should work on the whole healing processes for different wound types. Besides, it is necessary to develop stimuli-responsive wound dressings that are sensitive to the pH, temperature, moisture or infection of wounds, and trigger drug release under the environmental stimuli, which possess better prospects for clinical applications.

(3) In order to meet the mechanical properties of load-bearing soft tissues, cellulose-based hydrogels reinforced with inorganic NPs have been widely developed. It is necessary to systematically evaluate the potential toxicity and the ultimate clearance mechanism of these NPs *in vivo*. In addition, by loading unique therapeutic agents such as black phosphorus (BP) nanosheets, cellulose-based hydrogels may be explored for photothermal therapy (PTT) of bone disease and subsequent bone regeneration, since the intrinsic physiochemical properties of BP nanosheets endow them with excellent *in situ* biomineralization performance to facilitate osseointegration.¹³⁴

(4) Fluorescent hydrogels with unique properties such as injectability, fluorescence and safety, possess great potential in bioimaging applications. By embedding fluorescent materials (*e.g.* QDs, PP, and carbon dots) in cellulose matrices, several cellulose-based hydrogels have been successfully prepared and applied in FLI, which showed promise for biomedical fields and forensic medicine. Looking to the future, more cellulose-based hydrogels with unique properties should be explored, and the integration of bioimaging and treatment into a single hydrogel system is expected to realize the real-time and non-invasive visual *in vivo* degradation of injectable hydrogels, which possess great potential for image guided therapies.

(5) Self-healing hydrogels have attracted significant attention for the development of smart wearable devices. However, it still remains a great challenge to develop an integrated conductive cellulose-based hydrogel combining excellent self-healing and mechanical properties, which are two contradictory characteristics. Besides, to achieve ultrasensitive detection, (i) the hydrogels must be self-healed in shorter time, and the development of mechanically robust hydrogels with autonomous self-healing on the time scale of seconds is highly desirable; (ii) the hydrogels must respond to damage autonomously, rapidly and repeatedly, and respond to damage from the micro-to macroscale; and (iii) the hydrogels must retain their original mechanical and rheological properties after self-healing. Although the development of self-healing cellulose-based hydrogel sensors is still in its infancy, many positive explorations have been conducted and some exciting results have been achieved, such as the PVA-PVP-CNCs-Fe³⁺ hydrogel²⁹ and PAA-TA@CNCs-Al³⁺ hydrogels³⁰ exhibiting great potential for human activity monitoring and personal healthcare diagnosis.

(6) Except for the potential applications of cellulose-based hydrogels in biomedical fields including drug delivery, tissue engineering, wound dressing, bioimaging, and wearable sensors, more attention should be paid to other areas such as agriculture. The utilization of pesticides and chemical fertilizers, and water contamination can directly affect food safety, which is closely related to human health. Although great efforts have been made in wastewater treatment by using cellulose-based hydrogels to adsorb heavy metals and dye pollutants,^{44,49,135–137} there is still a chance to utilize cellulose-based hydrogels to realize the controlled release of chemical fertilizers,¹³⁸ which is expected to ensure production and reduce the usage of fertilizers.

Abbreviations

3D	Three-dimensional
4a-Phe	4-Amino-DL-phenylalanine
AA	Acetic acid
BC	Bacterial cellulose
BSA	Bovine serum albumin
CMC	Carboxymethyl cellulose
CNCs	Cellulose nanocrystals
CEC-TPH	Carboxyethyl cellulose-g-dithiodipropionate dihydrazide
CNW	Cellulose nanowhisker
DOX	Doxorubicin
EB	Electron beam
ECH	Epichlorohydrin
FLI	Fluorescence imaging
G''	Loss modulus
G'	Storage modulus
HA	Hyaluronic acid
HAp	Hydroxyapatite
HE	Healing efficiency
HEC	Hydroxyethyl cellulose
HPC	Hydroxypropyl cellulose

HPMC	Hydroxypropylmethyl cellulose
LCST	Lower critical solution temperature
MBA	<i>N,N'</i> -Methylenebisacrylamide
MC	Methyl cellulose
NPs	Nanoparticles
PAA	Poly(acrylic acid)
PAAAm	Poly(acrylic acid- <i>co</i> -acrylamide)
PAAm	Polyacrylamide
PBS	Phosphate buffer solution
PDEAAm	Poly(<i>N,N</i> -diethylacrylamide)
PDMAEMA	Poly(<i>N,N</i> -dimethylaminoethyl methacrylate)
PEG	Poly(ethylene glycol)
PEG-DA	Dibenzaldehyde-terminated poly(ethylene glycol)
PMVE	Poly(methyl vinyl ether)
PNIPAAm	Poly(<i>N</i> -isopropylacrylamide)
PP	Phosphor
PVA	Poly(vinyl alcohol)
PVCL	Poly(<i>N</i> -vinylcaprolactam)
PVP	Poly(vinylpyrrolidone)
QC	Quaternized cellulose
QDs	Quantum dots
SA	Sodium alginate
TA	Tannic acid
UCST	Upper critical solution temperature
UV	Ultraviolet

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

Financial support from the National Natural Science Foundation of China (21807074 and 51802202), the Guangdong Province Natural Science Foundation of PhD Start-up Fund (2018A030310574 and 2018A030310566), the Postdoctoral Science Foundation of China (2018M633123 and 2017M612727), the Fundamental Research Funds for the Central Universities (2017ZY49), and the National Postdoctoral Program for Innovative Talents (BX201600111) is gratefully acknowledged.

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